

Boundedness of Good Minimal Models

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1 Background

- Classification of algebraic varieties
- Boundedness and moduli

2 Main results

- Failure of boundedness for polarized lc good minimal models
- Boundedness of polarized klt good minimal models
- Moduli of polarized klt good minimal models

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Classification of curves

One of the central problems in algebraic geometry is to classify algebraic varieties up to birational equivalence.

For smooth projective curves C , the classification is determined by the sign of the canonical divisor K_C :

Type	Condition on K_C	Geometry
Fano	$-K_C$ ample	$\mathbb{P}^1$
Calabi–Yau	$K_C \sim 0$	elliptic curves
General type	K_C ample	genus $g \geq 2$

Surfaces and Kodaira dimension

For a smooth projective surface Y , the **Kodaira dimension** is defined by

$$\kappa(Y) := \dim \operatorname{Im}(\phi|_{mK_Y}) \quad (m \gg 0),$$

or equivalently, the growth rate of $h^0(Y, mK_Y)$.

After running the Minimal Model Program $Y \dashrightarrow X$, we obtain either a **minimal model** or a **Mori fiber space**:

$\kappa(Y)$	Output of MMP (on X)
$-\infty$	Mori fiber space: $\mathbb{P}^2$ or $\mathbb{P}^1$ -fibration
0	$K_X \sim_{\mathbb{Q}} 0$: K3, Enriques, abelian, bielliptic
1	K_X semiample: elliptic fibration over a curve
2	K_X nef and big: X admits a canonical model Z with K_Z ample

The Minimal Model Program (MMP) and Abundance Conjecture (completely established in dimension ≤ 3) predict that any projective variety Y with mild singularities is birational to either:

- a **Mori fiber space** $X \rightarrow Z$, i.e. $\rho(X/Z) = 1$, $-K_X$ is ample over Z (a Fano fibration), and $\dim X > \dim Z$;
- a **good minimal model** X , i.e. K_X is semiample and defines a Calabi–Yau fibration $X \rightarrow Z$.

Therefore, **Fano varieties**, **Calabi–Yau varieties**, and **canonically polarized varieties**, together with their iterated **fibrations**, form the basic building blocks in birational geometry.

Moduli spaces: general idea

In algebraic geometry, many classification problems are better understood via **moduli spaces**.

Roughly speaking, a moduli space should parametrize isomorphism classes of varieties with fixed numerical invariants (e.g. dimension, volume).

Expected features:

- points correspond to varieties (up to isomorphism);
- families correspond to morphisms into the moduli space;
- degenerations are included via a suitable compactification.

A fundamental guiding example is the moduli space of curves.

Fix $g \geq 2$. The moduli stack $\mathcal{M}_g$ parametrizes smooth projective curves of genus g , and it admits a coarse moduli space M_g .

Key facts:

- M_g is not compact: smooth curves can degenerate.
- The Deligne–Mumford stack $\overline{\mathcal{M}}_g$ provides a natural compactification.
- It admits a coarse moduli space $\overline{M}_g$.
- Boundary points correspond to **stable curves** with nodal singularities.

Why singularities appear

In higher-dimensional birational geometry, singular varieties naturally appear for two reasons:

- **MMP operations:** flips and contractions typically produce singular varieties.
- **Moduli compactifications:** smooth varieties degenerate to singular ones.

Therefore, one must enlarge the category from smooth varieties to varieties with controlled singularities.

Mild singularities (klt, lc, ϵ -lc)

Let X be a normal variety such that K_X is $\mathbb{Q}$ -Cartier.

To measure singularities, we take a log resolution $\pi : X' \rightarrow X$ and write

$$K_{X'} = \pi^* K_X + \sum a_i E_i.$$

The numbers a_i measure how singular X is along E_i .

Definitions:

- klt: all $a_i > -1$
- lc: all $a_i \geq -1$
- ϵ -lc: all $a_i \geq -1 + \epsilon$

Remark: If you are not familiar with singularities, you may think of X as smooth.

The notion of **boundedness** naturally arises from the theory of moduli spaces. Roughly speaking, a moduli space requires that all objects in a given class are controlled by a **finite type** parameter space.

Definition (Boundedness)

Let $d \in \mathbb{N}$ and let $\mathcal{P}$ be a set of d -dimensional projective varieties. We say $\mathcal{P}$ is a **bounded family** if there exists $r \in \mathbb{N}$ such that for every $X \in \mathcal{P}$, there exists a **very ample** divisor H on X with $H^d \leq r$. Equivalently, there exists a projective morphism $\pi : \mathcal{X} \rightarrow S$ of **finite type** schemes such that every $X \in \mathcal{P}$ is isomorphic to a fiber $\mathcal{X}_s$.

Example (curves $g \geq 2$):

- K_C is ample,
- $3K_C$ is very ample and embeds $C \hookrightarrow \mathbb{P}^N$.

Higher-dimensional analogue of Deligne–Mumford (curves):

There exists a **projective coarse moduli space** for X such that:

- X is a d -dimensional projective variety,
- X has **slc** (non-normal analogue of lc) singularities,
- K_X is **ample** with fixed volume $(K_X)^d = v$.

This theory has been developed over the past three decades through the contributions of many people.

Key input: boundedness of such varieties.

Theorem (Hacon-McKernan-Xu 2018)

Fix $d \in \mathbb{N}$ and $v \in \mathbb{Q}^{>0}$. Consider the set of varieties X such that

- X is **slc** of dimension d ,
- K_X is **ample** with volume $(K_X)^d = v$.

Then X belongs to a bounded family.

A celebrated result is Birkar's solution of the **BAB conjecture**.

Theorem (Birkar 2021)

Fix $d \in \mathbb{N}$ and $\epsilon \in \mathbb{Q}^{>0}$. Consider the set of varieties X such that

- X is ϵ -**lc** of dimension d ,
- $-K_X$ is **ample**.

Then X belongs to a bounded family.

For Calabi–Yau varieties, there is in general no natural choice of polarization. As a consequence, they are **not bounded** in the category of algebraic varieties.

For example, projective K3 surfaces and abelian varieties (of any fixed dimension) are not bounded.

It is **widely open** whether strict Calabi–Yau manifolds of dimension $d \geq 3$ are bounded.

Definition (strict Calabi–Yau). A smooth projective variety X is strict Calabi–Yau if

- X is simply connected,
- $K_X \sim 0$,
- $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < \dim X$.

Boundedness of polarized Calabi-Yau varieties

When studying moduli of smooth Calabi-Yau varieties, one may choose a **line bundle** as the polarization.

Since such moduli spaces are **not proper**, one needs to allow singular degenerations. In this setting, the limiting polarization is **no longer Cartier**, but only a $\mathbb{Q}$ -Cartier **Weil divisor**.

Theorem (Birkar 2023)

*Let $d \in \mathbb{N}$ and $u \in \mathbb{Q}^{>0}$. Consider Calabi-Yau varieties X and $\mathbb{Q}$ -Cartier **Weil** divisors A on X such that*

- X is **klt** of dimension d ,*
- A is an **ample** divisor on X such that $\text{vol}(A) \leq u$,*

then X belongs to a bounded family.

In particular, the Cartier index of A is bounded.

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Faiture of boundedness for polarized lc good minimal models

When X is a **good minimal model**, that is, K_X is semiample, there exists a CY fibration $f: X \rightarrow Z$ to a normal variety Z with $\dim Z = \kappa(X)$ and a canonical bundle formula

$$K_X \sim_{\mathbb{Q}} f^*(K_Z + B_Z + M_Z)$$

such that $K_Z + B_Z + M_Z$ is ample. We can then regard $(Z, B_Z + M_Z)$ as a **generalised log canonical (glc) model**.

Theorem (Hacon–J. 2025)

*There exist projective **lc** minimal **surfaces** X with $\kappa(X) \in \{0, 1, 2\}$ and ample $\mathbb{Q}$ -Cartier **Weil** divisors A such that $\text{vol}(K_X + A)$ is fixed, but the Cartier index of A is unbounded. Moreover, X is not bounded.*

Polarized klt good minimal models

Motivation. As a consequence of the **unboundedness** of polarized **lc** good minimal models, constructing **compact moduli spaces** for polarized Calabi–Yau varieties (and more generally good minimal models) becomes significantly more difficult.

Setup. We restrict to **polarized klt good minimal models** (X, A) :

- X is **klt**,
- K_X semiample, defining the Iitaka fibration $f: X \rightarrow Z$ with general fiber F ,
- A a $\mathbb{Q}$ -Cartier **Weil divisor** on X , **ample over** Z .

Key input (Birkar 2021, 2023).

- **Bases:** generalized log canonical models are bounded
- **Fibers:** polarized Calabi–Yau varieties are bounded

Boundedness of polarized klt good minimal models I

Theorem (J.-Jiao-Zhu 2025)

Let $d \in \mathbb{N}$, and $u, v \in \mathbb{Q}^{>0}$. Consider the set of polarized klt good minimal models $(X, A) \rightarrow Z$ satisfying the following:

- X is **klt** of dimension d ,
- A is **ample over** Z , with $\text{vol}(A|_F) \leq u$,
- $\text{Ivol}(K_X)(= (K_Z + B_Z + M_Z)^{\dim Z}) = v$.

Then the set of such X is boundedness in codimension one.

A set of varieties $\mathcal{P}$ is said to be **bounded in codimension one** if there exist a projective morphism $\pi : \mathcal{X} \rightarrow S$ between finite type schemes such that for every $X \in \mathcal{P}$ there exists $s \in S$ with X isomorphic to $\mathcal{X}_s$ in codimension one.

Boundedness of polarized klt good minimal models II

Theorem (J. 2023)

Let $d \in \mathbb{N}$, $u \in \mathbb{Q}^{>0}$, and $\sigma \in \mathbb{Q}[t]$ be a polynomial. Consider the set of polarized klt good minimal models $(X, A) \rightarrow Z$ satisfying the following:

- X is **klt** of dimension d ,
- A is **ample over** Z , with $\text{vol}(A|_F) \leq u$,
- $(K_X + tA)^d = \sigma(t)$.

Then the set of such X forms a bounded family. Moreover, there exist positive natural numbers m and τ , depending only on (d, u, σ) , such that $m(K_X + \tau A)$ is **very ample**.

We denote this set by $\mathcal{G}_{\text{klt}}(d, u, \sigma)$.

Moduli of polarized klt good minimal models

Theorem (J. 2023)

Fix $d \in \mathbb{N}$, $u \in \mathbb{Q}^{>0}$, and $\sigma \in \mathbb{Q}[t]$. The moduli functor $\mathfrak{G}_{\text{klt}}(d, u, \sigma)$ for the set $\mathcal{G}_{\text{klt}}(d, u, \sigma)$ is a separated Deligne–Mumford stack of finite type, which admits a coarse moduli space $G_{\text{klt}}(d, u, \sigma)$ as a separated algebraic space.

We prove that a certain line bundle is ample on $\tilde{G}_{\text{klt}}(d, u, \sigma)$.

Theorem (J. 2026)

Let $\delta : \tilde{G}_{\text{klt}}(d, u, \sigma) \rightarrow G_{\text{klt}}(d, u, \sigma)$ be the normalization of $G_{\text{klt}}(d, u, \sigma)$, then $\tilde{G}_{\text{klt}}(d, u, \sigma)$ is a quasi-projective scheme. Moreover, if the non-normal locus of $G_{\text{klt}}(d, u, \sigma)$ is proper, then $G_{\text{klt}}(d, u, \sigma)$ is a quasi-projective scheme.

There exists an example due to Kollár of a line bundle L on a non-normal, non-proper space X such that L is not ample, whereas its pullback to the normalization $\tilde{X}$ is ample. Therefore, the quasi-projectivity of $\tilde{G}_{\text{klt}}(d, u, \sigma)$ does not imply that of $G_{\text{klt}}(d, u, \sigma)$. 

Weak positivity

The quasi-projectivity result is based on Viehweg's ampleness criterion in the non-proper setting and the following weak positivity (a non-proper analogue of nefness) result.

Theorem (J. 2026)

Let $f: X \rightarrow S$ be a locally stable family of klt varieties over a reduced quasi-projective scheme S such that K_X is f -semiample. Let L be an f -semiample line bundle on X . Assume that:

- there exists $l \in \mathbb{N}$ such that $\text{lct}(X_s, |L_s|) > \frac{1}{l}$ for all $s \in S$, and
- the sheaf f_*L is locally free of rank $r > 0$ and compatible with arbitrary base change.

Then for every natural number $q \geq 2$ such that qK_X is Cartier, the sheaf

$$\left(\bigotimes^r f_*(\mathcal{O}_X(qK_{X/S}) \otimes L^q) \right) \otimes \det(f_*L)^{-q}$$

is weakly positive over S .

Thank you!

I would like to thank my supervisor for guidance and the committee members for their participation.

Questions are welcome.